

Neutron-star universal relations from equation-of-state response geometry

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Neutron-star observables obey quasi-universal relations despite their strong individual dependence on the equation of state (EOS). We formulate this behavior as a property of the differential map from EOS function space to relativistic stellar observables. For observable coordinates y^A , EOS coordinates a^i , and a declared covariance C_{EOS}^{ij} on EOS perturbations, we study the induced response $G_{\text{EOS}}^{AB} = J^A_i C_{\text{EOS}}^{ij} J^B_j$, with $J^A_i = \partial y^A / \partial a^i$, after quotienting displacement along a stellar sequence. In the space $(\ln C, \ln \bar{I}, \ln \Lambda_2)$, the resulting normal response contains a strongly suppressed mode that is recovered without imposing an empirical I–Love fit. Across a broad causal latent-sound-speed ensemble, the median soft-to-hard normal RMS ratio is 0.020, while the soft covector has median cosine alignment 0.998 with the I–Love normal. The soft one-form is approximately integrable: a high-resolution Frobenius diagnostic is $\lesssim 7.4 \times 10^{-3}$ over the tested domain, and a scalar potential reconstructed only from local response gradients generalizes across held-out EOS backgrounds. Second-order curvature predicts finite normal displacements with median relative error 2.4% at latent perturbation amplitude 0.1, but loses quantitative accuracy for order-unity deformations. Finally, localized sound-speed softening can strongly degrade the ordinary I–Love projection while preserving the full soft-mode hierarchy; the added sensitivity is concentrated on the softened layer. These results separate quasi-universality into local EOS insensitivity, integrability, and nonlinear curvature, and provide a differential description of both its persistence and controlled breakdown.

I. INTRODUCTION

A neutron star maps an equation of state into a collection of exterior and bulk observables through the relativistic stellar-structure equations. Individual observables such as the radius, moment of inertia, and tidal deformability retain substantial dependence on the EOS. Certain combinations, however, vary much less. The I–Love–Q relations provide the canonical example, relating the dimensionless moment of inertia, tidal deformability, and spin-induced quadrupole moment with weak EOS dependence [1]. Their accuracy has motivated both practical applications and a long-running question: why does eliminating the stellar-configuration variable remove so much of the EOS dependence?

Several complementary explanations already exist and set the starting point for this work. Approximate self-similarity of isodensity contours and the dominance of the outer core were identified as important ingredients [2]. The proximity of realistic compact stars to stiff or incompressible configurations provides another perspective [3]. Perturbative calculations sharpened this statement: the I–Love relation can be stationary with respect to EOS variations around the incompressible limit [4], and in Newtonian gravity the stationarity of moment–Love relations extends to arbitrary density perturbations about an incompressible star [5]. More recently, Hu, Gao, and Shao separated linear deviations of the I–C and I–Love relations into an EOS-difference factor and a background-structure factor [6], while Kamata, Minamiguchi, and Minato derived asymptotic universal relations directly from the relativistic stellar equations and boundary conditions [7]. These results explain or quantify why already identified relations are insensitive in controlled limits and provide important first-order and asymptotic antecedents for the construction below.

The modern EOS literature also makes clear that the term universal is conditional. Flexible nonparametric EOS ensembles produce larger deviations than compact catalogs of named EOSs and expose relations whose apparent accuracy depends on the EOS prior [8]. Statistical and machine-learning methods have been used to search directly for combinations of observables with small EOS scatter [9, 10]. Automatic differentiation through the stellar-structure problem is practical [11], while sensitivity matrices have been used to determine which microscopic EOS parameters most strongly affect neutron-star observables [12].

These developments suggest a different formulation. Instead of beginning with a proposed relation and measuring its scatter, we begin with the differential map from EOS function space to observable space. We ask which normal directions in observable space are weakly excited by physically weighted EOS perturbations after displacement along the stellar sequence has been removed. This turns quasi-universality into a local geometric statement and separates three questions that are usually combined:

1. Does the EOS response possess a small transverse direction locally?
2. Do those local directions integrate to a global relation?
3. What controls the finite departure from that relation?

The distinction matters. A small local response need not define a global function $F(\mathbf{y})$, and a first-order stationary relation can still acquire finite scatter from second-order curvature. Conversely, a breakdown of universality may be attributable to a specific density-localized EOS direction rather than to an undifferentiated increase in catalog scatter.

This paper develops that construction for cold, barotropic neutron stars in general relativity. We initially restrict the observable space to

$$\mathbf{y} = (\ln C, \ln \bar{I}, \ln \Lambda_2), \quad C \equiv M/R, \quad \bar{I} \equiv I/M^3, \quad (1)$$

and treat the known I–Love relation as a stringent validation target. Compactness–Love provides a deliberately weaker control. We do not assume either relation when constructing the response geometry.

The calculation recovers a strongly separated soft normal mode without using an empirical I–Love fit, and that mode remains nearly parallel to the I–Love normal over broad causal EOS deformations. The resulting one-form is approximately integrable, while second-order curvature controls the leading finite departure from the local soft tangent plane. A controlled localized sound-speed softening experiment then shows that the I–Love projection can degrade substantially while the full low-response hierarchy survives through a rotation of the soft covector. The extra sensitivity can be localized to the responsible stellar layer.

Section II defines the EOS response operator, the induced metric in observable space, the stellar-sequence quotient, integrability, and the second-order curvature terms. Section III specifies the stellar-structure problem and EOS perturbations. Section IV gives the numerical validation. Section V presents the local response hierarchy and full normal spectrum, Sec. VI tests integrability and reconstructs the scalar potential, Sec. VII tests the finite-radius curvature expansion, and Sec. VIII isolates a controlled breakdown mechanism.

II. RESPONSE GEOMETRY

A. The stellar map

Let $\mathcal{E}$ denote a cold barotropic EOS and let h_c denote the central relativistic enthalpy labeling a stellar configuration. Solving the stellar-structure and perturbation equations defines

$$\Phi : (\mathcal{E}, h_c) \longrightarrow \mathbf{y}. \quad (2)$$

The EOS is an infinite-dimensional object. For numerical work we introduce coordinates a^i on a finite approximation to EOS function space, with the functional limit recovered under basis refinement.

The differential of Eq. (2) separates into an EOS part and a stellar-sequence part,

$$dy^A = J^A_i da^i + t^A dh_c, \quad (3)$$

where

$$J^A_i \equiv \left. \frac{\partial y^A}{\partial a^i} \right|_{h_c}, \quad t^A \equiv \left. \frac{\partial y^A}{\partial h_c} \right|_{\mathcal{E}}. \quad (4)$$

The vector t^A is tangent to a fixed-EOS stellar sequence. Quasi-universality concerns the extent to which changing the EOS moves this sequence transversely rather than merely reparameterizing motion along it.

B. Metric-aware EOS response

A singular-value decomposition of J^A_i is not by itself a coordinate-independent characterization of EOS sensitivity. Under a change of EOS coordinates $a^i \rightarrow \tilde{a}^\alpha(a)$, the numerical scale of the columns of J changes. In particular, a rescaling $a^i \rightarrow \lambda_i a^i$ changes the singular spectrum without changing the underlying set of EOSs.

We therefore equip EOS perturbations with a positive covariance or inverse metric C_{EOS}^{ij} . In a probabilistic interpretation this can be the local covariance of a nonparametric EOS process; geometrically it defines which perturbations are considered comparable. The EOS-induced second moment in observable space is

$$G_{\text{EOS}}^{AB} = J^A_i C_{\text{EOS}}^{ij} J^B_j. \quad (5)$$

Under a consistent coordinate transformation of both J and C_{EOS} , Eq. (5) is invariant.

In the functional limit, choose an EOS field $s(h)$, for example a latent field controlling the sound speed, and define

$$K^A(h) = \frac{\delta y^A}{\delta s(h)}. \quad (6)$$

Then

$$G_{\text{EOS}}^{AB} = \int dh dh' K^A(h) C_{\text{EOS}}(h, h') K^B(h'). \quad (7)$$

Equation (7) is the central first-order object.

C. Quotienting the stellar-sequence direction

Observable space also requires a metric to define orthogonality and normalize covectors. Let g_{AB} denote a declared positive metric on observable space. A Euclidean metric in logarithmic observables is a useful baseline, but it is not assumed unique; coordinate dependence is a required robustness test.

The normalized sequence tangent is

$$\hat{t}^A = \frac{t^A}{(g_{BC} t^B t^C)^{1/2}}. \quad (8)$$

The projector onto directions g -orthogonal to the sequence is

$$P^A{}_B = \delta^A{}_B - \hat{t}^A \hat{t}_B, \quad \hat{t}_B = g_{BC} \hat{t}^C. \quad (9)$$

The transverse EOS response is

$$G_{\perp}^{AB} = P^A{}_C G_{\text{EOS}}^{CD} P^B{}_D. \quad (10)$$

For a two-dimensional observable plane such as $(\ln \Lambda_2, \ln \bar{I})$, the quotient by the one-dimensional sequence tangent leaves a single normal direction. Its response gives a local scalar measure of non-universality. In higher-dimensional observable spaces, the physical hierarchy is obtained from the constrained cotangent-space problem below, equivalently a generalized eigenproblem normalized with g^{AB} ; raw Euclidean eigenvalues of PGP^T are not themselves coordinate invariant.

D. Local quasi-universal covectors

A covector n_A defines a local candidate relation when its EOS-induced variance

$$\mathcal{V}[n] = n_A G_{\text{EOS}}^{AB} n_B \quad (11)$$

is small subject to

$$n_A t^A = 0, \quad n_A g^{AB} n_B = 1. \quad (12)$$

Equivalently, n_A is a low-response mode in the normal cotangent space to the stellar sequence. This definition makes prior dependence explicit through C_{EOS} , rather than hiding it in the composition of a finite EOS catalog.

E. From local insensitivity to a global relation

A local normal field is not automatically the gradient of a scalar relation. A global quasi-universal relation requires a function $F(\mathbf{y})$ such that

$$F(\mathbf{y}) \simeq \text{const}, \quad n_A(\mathbf{y}) \propto \partial_A F. \quad (13)$$

Thus the existence of a suppressed local response mode and the existence of a global fitted curve or hypersurface are distinct statements.

For a codimension-one distribution represented by the one-form

$$\omega = n_A dy^A, \quad (14)$$

Frobenius integrability requires

$$\omega \wedge d\omega = 0 \quad (15)$$

where the condition is nontrivial. In the two-observable I–Love plane local line fields are automatically integrable, but path dependence becomes nontrivial when the construction is embedded in a larger observable/configuration space or when several low-response directions are retained. We therefore use both local Frobenius diagnostics in higher dimension and direct path-dependence tests of transported covectors.

Operationally, the global relation is reconstructed after determining n_A from the stellar response. The standard empirical I–Love fit is used only for comparison.

F. Second order and residual scatter

Suppose $F(\Phi)$ is stationary at first order under EOS perturbations. For finite EOS displacement δa^i ,

$$\begin{aligned} \delta F &= \partial_A F J^A_i \delta a^i \\ &+ \frac{1}{2} [\partial_A F H^A_{ij} + \partial_A \partial_B F J^A_i J^B_j] \delta a^i \delta a^j + \mathcal{O}(\delta a^3), \end{aligned} \quad (16)$$

with

$$H^A_{ij} = \frac{\partial^2 y^A}{\partial a^i \partial a^j}. \quad (17)$$

When the first line of Eq. (16) is suppressed, the bracketed term is the leading local source of finite variation in the scalar relation. It contains two conceptually distinct pieces: the curvature H^A_{ij} of the EOS-to-observable map and the curvature $\partial_A \partial_B F$ of the chosen relation in observable space. In Sec. VII we test the first piece directly by freezing the local soft normal and measuring displacement away from its tangent plane; we do not claim there to have separately validated the complete second-order expansion of the globally reconstructed F .

G. Relation to previous stationarity results

The perturbative results of Refs. [4, 5] imply a vanishing or strongly suppressed first variation for specific moment–Love combinations near the incompressible limit. In the present notation, such stationarity corresponds schematically to

$$n_A J^A_i \simeq 0 \quad (18)$$

for the relevant perturbations. The difference here is the order of construction. We do not begin with a chosen n_A . We calculate the EOS response of several observables, quotient the sequence direction, and infer the low-response normal field from the resulting operator. The known I–Love relation is therefore a test of the construction.

III. RELATIVISTIC STELLAR STRUCTURE

A. Background configurations

For a static spherical star,

$$ds^2 = -e^{2\nu(r)} dt^2 + \left(1 - \frac{2m(r)}{r}\right)^{-1} dr^2 + r^2 d\Omega^2. \quad (19)$$

The Tolman–Oppenheimer–Volkoff equations are

$$\frac{dm}{dr} = 4\pi r^2 \epsilon, \quad (20)$$

$$\frac{dp}{dr} = -\frac{(\epsilon + p)(m + 4\pi r^3 p)}{r(r - 2m)}. \quad (21)$$

We use relativistic enthalpy

$$h(p) = \int_0^p \frac{dp'}{\epsilon(p') + p'} \quad (22)$$

as the integration coordinate. Since

$$\frac{dh}{dr} = -\frac{m + 4\pi r^3 p}{r(r - 2m)}, \quad (23)$$

the background system integrated in the code is

$$\frac{dr}{dh} = -\frac{r(r - 2m)}{m + 4\pi r^3 p}, \quad (24)$$

$$\frac{dm}{dh} = 4\pi r^2 \epsilon \frac{dr}{dh}. \quad (25)$$

The stellar surface is at $h = 0$, and

$$M = m(R), \quad C = M/R. \quad (26)$$

The enthalpy equations are singular at the exact center. We start a small interval Δh below h_c using the regular series

$$r_0^2 = \frac{3\Delta h}{2\pi(\epsilon_c + 3p_c)} + \mathcal{O}(\Delta h^2), \quad (27)$$

$$m_0 = \frac{4\pi}{3}\epsilon_c r_0^3 + \mathcal{O}(r_0^5). \quad (28)$$

The production fixed-step integration presently uses $\Delta h/h_c = 10^{-4}$; convergence with respect to both this offset and the number of integration steps is part of the validation described in Sec. IV.

B. Quadrupolar tidal response

We use the standard relativistic static $l = 2$ tidal equations [13, 14]. Defining

$$y(r) \equiv r \frac{H'(r)}{H(r)}, \quad (29)$$

the first-order equation is

$$r \frac{dy}{dr} + y^2 + F(r)y + r^2 Q(r) = 0, \quad (30)$$

with

$$F(r) = \frac{1 - 4\pi r^2(\epsilon - p)}{1 - 2m/r}, \quad (31)$$

and

$$Q(r) = \frac{4\pi}{1 - 2m/r} \left[5\epsilon + 9p + \frac{\epsilon + p}{c_s^2} - \frac{6}{4\pi r^2} \right] - 4 \left[\frac{m + 4\pi r^3 p}{r^2(1 - 2m/r)} \right]^2. \quad (32)$$

In enthalpy coordinates we use $dy/dh = (dy/dr)(dr/dh)$, with regular central condition

$$y(r \rightarrow 0) = 2 + \mathcal{O}(r^2). \quad (33)$$

For an EOS with nonzero surface energy density, the logarithmic derivative must include the surface-density jump [14],

$$y_R = y(R^-) - \frac{4\pi R^3 \epsilon_s}{M}. \quad (34)$$

The Love number is then

$$\begin{aligned} k_2 = & \frac{8C^5}{5}(1-2C)^2 [2 + 2C(y_R - 1) - y_R] \\ & \times \left\{ 2C[6 - 3y_R + 3C(5y_R - 8)] \right. \\ & + 4C^3[13 - 11y_R + C(3y_R - 2) + 2C^2(1 + y_R)] \\ & \left. + 3(1-2C)^2[2 - y_R + 2C(y_R - 1)] \ln(1-2C) \right\}^{-1}. \end{aligned} \quad (35)$$

We report the dimensionless tidal deformability

$$\Lambda_2 = \frac{2}{3} k_2 C^{-5}. \quad (36)$$

C. Moment of inertia

The moment of inertia is calculated at first order in uniform rotation using Hartle's slow-rotation formalism [15]. Let $\bar{\omega}(r) = \Omega - \omega(r)$ and

$$j(r) = e^{-\nu(r)} \sqrt{1 - \frac{2m(r)}{r}}. \quad (37)$$

The frame-dragging equation is

$$\frac{d}{dr} \left(r^4 j \frac{d\bar{\omega}}{dr} \right) + 4r^3 \frac{dj}{dr} \bar{\omega} = 0. \quad (38)$$

Only $d \ln j / dr$ is needed in the first-order system, so the arbitrary normalization of ν never enters. Near the center,

$$\bar{\omega}(r) = 1 + \frac{8\pi}{5} (\epsilon_c + p_c) r^2 + \mathcal{O}(r^4), \quad (39)$$

where the overall normalization is arbitrary.

Matching to

$$\bar{\omega}_{\text{ext}}(r) = \Omega - \frac{2J}{r^3} \quad (40)$$

gives

$$J = \frac{R^4}{6} \bar{\omega}'(R), \quad \Omega = \bar{\omega}(R) + \frac{R}{3} \bar{\omega}'(R), \quad (41)$$

and therefore

$$I = \frac{R^4 \bar{\omega}'(R)}{6\bar{\omega}(R) + 2R\bar{\omega}'(R)}, \quad \bar{I} = \frac{I}{M^3}. \quad (42)$$

D. EOS coordinates and perturbations

The final response calculation requires EOS perturbations that approach a functional description while satisfying thermodynamic stability and causality by construction. The baseline representation uses a latent field $u(h)$ mapped to the sound speed,

$$c_s^2(h) = \sigma[u(h)], \quad \sigma(u) = \frac{1}{1 + e^{-u}}, \quad (43)$$

with stricter bounds applied explicitly if required. The latent field is expanded as

$$u(h) = u_0(h) + \sum_{i=1}^{N_b} a_i \phi_i(h). \quad (44)$$

The basis ϕ_i is a numerical representation only. Physical conclusions must converge as N_b and the basis family are varied.

For the initial response calculation, $u_0(h)$ is taken from a causal reference EOS above a matching enthalpy h_{match} . Below the match, the reference EOS is retained unchanged. Above it, the perturbed sound speed is used to reconstruct the thermodynamics rather than perturbing pressure and energy-density tables independently. From the definition of enthalpy and $c_s^2 = dp/d\epsilon$,

$$\frac{dp}{dh} = \epsilon + p, \quad (45)$$

$$\frac{d\epsilon}{dh} = \frac{\epsilon + p}{c_s^2(h)}. \quad (46)$$

Equations (45)–(46) are initialized with $p(h_{\text{match}})$ and $\epsilon(h_{\text{match}})$ from the reference EOS. Thus every finite perturbation used in the Jacobian corresponds to a thermodynamically consistent barotrope, with $0 < c_s^2 < 1$ enforced by the latent logistic map.

For the production response coordinates we use nodal values of the latent perturbation,

$$a_i \equiv \delta u(h_i), \quad u(h) = u_0(h) + \delta u(h), \quad (47)$$

with $\delta u(h)$ linearly interpolated between the nodes. This choice is not intended to assign physical meaning to individual nodes. Its purpose is to make basis refinement well defined: if $C_{\text{EOS}}(h, h')$ is a target functional covariance, then the finite-dimensional metric is simply its evaluation on the nodes,

$$C_{\text{EOS}}^{ij} = C_{\text{EOS}}(h_i, h_j). \quad (48)$$

Increasing the number of nodes therefore refines the representation of one fixed stochastic process rather than changing the normalization or meaning of the perturbation coefficients.

Our baseline target metric is the squared-exponential kernel

$$C_{\text{EOS}}(h, h') = \sigma_u^2 \exp \left[-\frac{(h - h')^2}{2\ell_h^2} \right]. \quad (49)$$

We also use Matérn-3/2 and exponential kernels to test sensitivity to rougher EOS perturbations. The amplitude σ_u scales the absolute linear-response variance but cancels from ratios of response scales formed with the same EOS metric. The correlation length ℓ_h , in contrast, changes which density scales are allowed to vary coherently and is therefore a physical robustness parameter.

For Stage 1 validation we additionally use two analytic EOS families. A constant-energy-density star provides the exact Schwarzschild-interior background solution. A relativistic $\Gamma = 2$ polytrope supplies the Newtonian $n = 1$ limits of k_2 and $I/(MR^2)$. Neither benchmark EOS defines the final EOS metric. The first response calculations use smooth causal relativistic polytropes only as controlled reference points; broad nonparametric EOS ensembles enter after derivative, node-refinement, and metric-robustness tests.

IV. NUMERICAL IMPLEMENTATION AND VALIDATION

The production stellar calculation uses a fixed-step fourth-order Runge–Kutta integration in relativistic enthalpy. Fixed control flow keeps the EOS-to-observable map differentiable and makes the derivative calculation transparent.

TABLE I. Relative error of the enthalpy-coordinate background solver against the exact constant-energy-density solution.

h_c	$\Delta M/M$	$\Delta R/R$	$\Delta C/C$
0.05	-2.16×10^{-6}	-7.23×10^{-7}	-1.44×10^{-6}
0.15	-1.19×10^{-6}	-4.00×10^{-7}	-7.93×10^{-7}
0.30	-5.57×10^{-7}	-1.88×10^{-7}	-3.70×10^{-7}

The background, static $l = 2$ tidal perturbation, and first-order frame-dragging equations are evolved together. An independent adaptive calculation in the radius coordinate, using DOP853, is retained only for validation.

Automatic differentiation is used to evaluate the finite-dimensional Jacobian J^A_i . It is a computational method rather than a validation. Every production derivative is checked against symmetric finite differences,

$$\frac{\partial y^A}{\partial a_i} \simeq \frac{y^A(a_i + \Delta a_i) - y^A(a_i - \Delta a_i)}{2\Delta a_i}, \quad (50)$$

over a range of step sizes.

A. Stellar-structure validation

The background integrator is first checked against the exact Schwarzschild-interior solution for a constant-energy-density star. Using 8192 enthalpy steps and the regular center offset $\Delta h/h_c = 10^{-4}$, the relative errors in M , R , and C are at the 10^{-6} level over the three central enthalpies listed in Table I.

The full observable solver is then compared against an independently coded radius-coordinate adaptive integration. This cross-check shares the physical equations but not the integration variable or numerical stepping. For a constant-energy-density configuration at $h_c = 0.15$, the maximum absolute relative difference among

$$(M, R, C, k_2, \Lambda_2, I, \bar{I}, I/MR^2)$$

is 4.71×10^{-7} . For a relativistic $\Gamma = 2$ polytrope at $h_c = 0.10$, the corresponding maximum difference is 8.10×10^{-7} . The largest discrepancy in both comparisons is in Λ_2 , whose C^{-5} dependence amplifies otherwise smaller differences in the background solution.

As a separate low-compactness check, the homogeneous-star solution at $C = 4.96 \times 10^{-3}$ gives

$$k_2 = 0.72897, \quad I/(MR^2) = 0.40171,$$

approaching the Newtonian values $3/4$ and $2/5$, respectively. A relativistic $\Gamma = 2$ polytrope at $C = 2.49 \times 10^{-3}$ gives

$$k_2 = 0.25543, \quad I/(MR^2) = 0.26206,$$

compared with the Newtonian $n = 1$ values

$$k_2 = \frac{15 - \pi^2}{2\pi^2} = 0.25991, \quad \frac{I}{MR^2} = \frac{2}{3} \frac{\pi^2 - 6}{\pi^2} = 0.26138.$$

These finite-compactness differences have the expected sign and scale and vanish as the central enthalpy is lowered.

B. Derivative and response validation

The response calculation is accepted only after the following additional tests:

1. automatic derivatives agree with symmetric finite differences;
2. the induced response converges as the nodal latent-field grid is refined while holding the target functional covariance fixed;
3. the I-Love versus C-Love hierarchy is repeated with more than one EOS covariance kernel and correlation length;

4. the hierarchy is repeated about more than one reference EOS before moving to a broad nonparametric ensemble;
5. local response scalars are invariant under observable-coordinate transformations when the observable metric is transformed consistently;
6. the result is stable to the enthalpy integration and EOS reconstruction resolution.

The observable-coordinate test is important because a numerical measure of thinness can change under reparameterization if the normal covector is normalized inconsistently. We therefore transform G^{AB} , t^A , and g_{AB} as tensors and covariant metrics, rather than comparing Euclidean norms in unrelated coordinate systems.

The initial Gaussian-mode diagnostic used in Sec. V predates the nodal production representation. It is retained as a reproducibility check only. All robustness and later science results use nodal values of the latent sound-speed field, for which grid refinement corresponds to a fixed functional EOS covariance.

V. LOCAL RESPONSE HIERARCHY

The first central test compares the EOS response transverse to the stellar sequence in the I-Love and C-Love planes. Both are constructed from the same stellar configurations and EOS metric, and no empirical relation is used to define either normal direction.

For each reference configuration we compute the covariance-weighted normal response. In a two-observable plane the normal cotangent direction is unique once its normalization metric is declared. If G^{AB} is the EOS-induced covariance, t^A the sequence tangent, and g_{AB} the observable metric, the scalar response is

$$\sigma_{\perp}^2 = n_A G^{AB} n_B, \quad n_A t^A = 0, \quad n_A g^{AB} n_B = 1. \quad (51)$$

The same EOS covariance is used when comparing I-Love and C-Love.

A. Development diagnostic

Before constructing a broad EOS ensemble, we performed a deliberately restricted diagnostic to determine whether the proposed response geometry contains the expected signal at all. The reference model is a relativistic $\Gamma = 2$ polytrope with $K = 100$ in geometrized units. Five Gaussian latent sound-speed modes span $h \in [0.02, 0.55]$. Their coefficient covariance is Eq. (49) with $\sigma_a = 0.1$ and $\ell_h = 0.08$. We use the Euclidean metric in $(\ln C, \ln \bar{I}, \ln \Lambda_2)$ for this diagnostic only.

Table II reports the resulting local normal RMS responses. Across the tested sequence, the I-Love normal response is smaller than the C-Love response by a factor between 7.5 and 10.8 in RMS, or roughly 57–116 in variance. This calculation uses one reference EOS, one EOS covariance, one basis resolution, and one observable metric and therefore is not yet evidence for prior-independent universality. Its purpose is narrower: the differential construction recovers a strong hierarchy without inserting an empirical I-Love fit.

At $h_c = 0.16$, the automatically differentiated observable Jacobian was compared component by component with a symmetric finite difference using a coefficient step 2×10^{-4} . The maximum relative discrepancy was 8.2×10^{-8} , with an RMS relative discrepancy 2.2×10^{-8} . This derivative check is specific to the diagnostic configuration; the production analysis repeats it under basis, integration, and EOS changes.

TABLE II. Restricted response-geometry diagnostic for the $\Gamma = 2$ reference sequence. The quoted RMS values are local first-order responses in logarithmic observable coordinates under the declared latent-field covariance.

h_c	C	$\sigma_{\perp}^{I\text{Love}}$	$\sigma_{\perp}^{C\text{Love}}$	$\sigma_{\perp}^{I\text{Love}}/\sigma_{\perp}^{C\text{Love}}$
0.08	0.0673	0.00198	0.02133	0.0929
0.12	0.0933	0.00254	0.02612	0.0974
0.16	0.1154	0.00297	0.02857	0.1040
0.20	0.1344	0.00335	0.03007	0.1115
0.25	0.1545	0.00382	0.03124	0.1223
0.30	0.1713	0.00425	0.03202	0.1328

TABLE III. Summary of the production local-response robustness tests. The ranges quote the I-Love/C-Love normal RMS ratio.

Test	RMS-ratio result
Nodal refinement, $N = 17 \rightarrow 33$	fractional change 3.6×10^{-5}
SE/Matérn-3/2/exponential, $\ell_h = 0.04\text{--}0.16$	0.1024–0.1034
I-weighted observable metric	0.1829–0.1847
Love-weighted observable metric	0.1048–0.1058
$\Gamma = 1.70\text{--}2.00$ reference sequences	0.096–0.207
1024/512 to 2048/1024 numerical resolution	fractional change 3.4×10^{-3}

B. Production nodal response and robustness

We next repeat the calculation using nodal latent-field values, Eq. (47), so that increasing the number of EOS coordinates refines a fixed functional covariance. At $h_c = 0.16$ for the $\Gamma = 2$ reference sequence, the I-Love/C-Love normal RMS ratio is

$$0.10102, \quad 0.10286, \quad 0.10279, \quad 0.10278$$

for 5, 9, 17, and 33 nodes, respectively. The 17-node result differs from the 33-node value by 3.6×10^{-5} fractionally. We use 17 nodes for the remaining robustness tests.

For the Euclidean metric in logarithmic observables, changing the EOS covariance among squared-exponential, Matérn-3/2, and exponential kernels and varying $\ell_h = 0.04, 0.08, 0.16$ gives

$$0.1024 \leq \frac{\sigma_{\perp}^{I\text{Love}}}{\sigma_{\perp}^{C\text{Love}}} \leq 0.1034. \quad (52)$$

Thus the order-of-magnitude suppression is insensitive to both the correlation length and the differentiability of the allowed EOS perturbations in this controlled reference problem.

Changing the observable normalization metric changes the numerical response scale, as it must. A metric that weights the $\ln \bar{I}$ direction four times more strongly gives RMS ratios 0.183–0.185, while weighting the $\ln \Lambda_2$ direction gives 0.1048–0.1058. The hierarchy therefore persists under these deliberately nonuniform normalizations, although the numerical ratio itself is metric dependent. This should be distinguished from a coordinate transformation: transforming from $(\ln C, \ln \bar{I}, \ln \Lambda_2)$ to $(C, \bar{I}, \Lambda_2)$ while transforming G^{AB} , t^A , and g_{AB} consistently changes the I-Love and C-Love variances by only 8.4×10^{-14} and 9.4×10^{-16} fractionally, respectively.

The hierarchy is also present away from the $\Gamma = 2$ reference. For smooth causal relativistic polytropes with $\Gamma = 1.70, 1.85, 2.00$ and $h_c = 0.12, 0.20, 0.30$, spanning $C = 0.073\text{--}0.171$, we find

$$0.096 \leq \frac{\sigma_{\perp}^{I\text{Love}}}{\sigma_{\perp}^{C\text{Love}}} \leq 0.207. \quad (53)$$

Even the weakest case therefore retains a factor 4.8 suppression in RMS response.

For the nodal production representation, a nine-node Jacobian at $h_c = 0.16$ agrees with a symmetric finite difference of step 2×10^{-4} to 6.5×10^{-8} maximum relative error and 1.3×10^{-8} RMS relative error. Increasing the stellar integration and thermodynamic-reconstruction resolutions from 1024/512 to 2048/1024 steps changes the response ratio by 3.4×10^{-3} . The coarser 512/256 calculation differs from the finest result by about one percent.

These tests establish H1 and H2 for a controlled family of smooth causal reference EOSs: the known I-Love relation appears as a strongly suppressed transverse EOS response without entering the construction.

C. Full normal spectrum

The two-dimensional comparison above tests a chosen projection of the observable space. The response-geometry formulation makes a stronger prediction in the full three-dimensional space $(\ln C, \ln \bar{I}, \ln \Lambda_2)$. After quotienting the stellar-sequence tangent, two independent normal directions remain. We denote their EOS-induced variances by

$$\lambda_{\text{soft}} \leq \lambda_{\text{hard}}$$

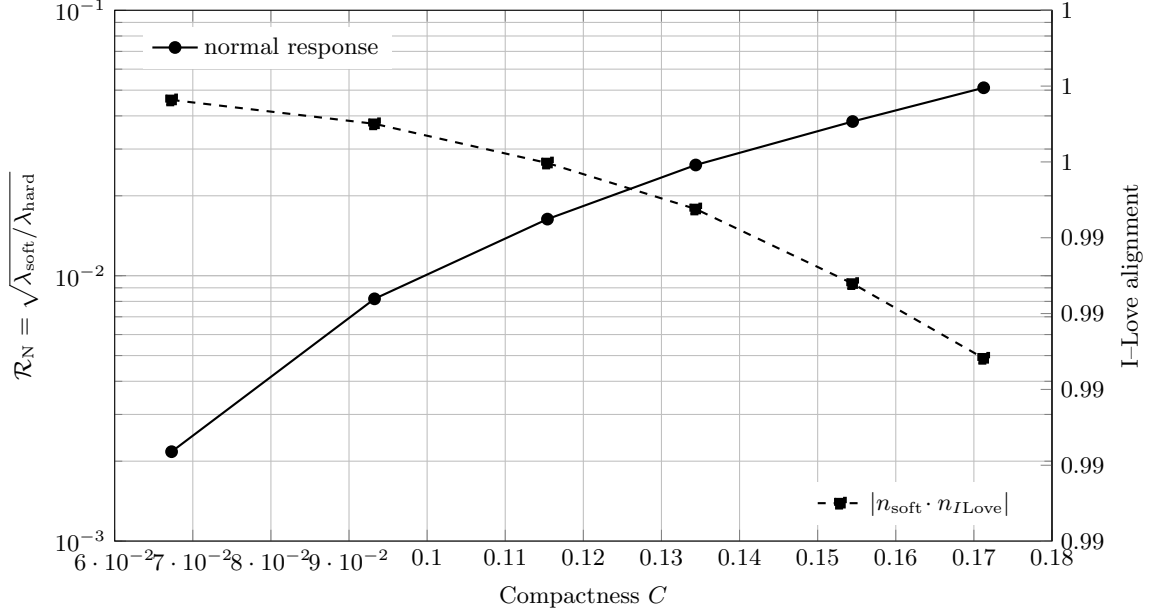


FIG. 1. Normal-response hierarchy along the $\Gamma = 2$ reference sequence. After quotienting the stellar-sequence tangent, the soft-to-hard normal RMS ratio remains well below unity (left axis), while the soft covector stays closely aligned with the embedded I-Love normal (right axis).

and characterize the hierarchy by

$$\mathcal{R}_N \equiv \sqrt{\frac{\lambda_{\text{soft}}}{\lambda_{\text{hard}}}}. \quad (54)$$

For the $\Gamma = 2$ reference sequence, $\mathcal{R}_N$ increases from 2.2×10^{-3} at $C = 0.067$ to 5.1×10^{-2} at $C = 0.171$. Thus the quotient response is itself strongly anisotropic: one normal direction is between roughly 20 and 460 times quieter in RMS than the other over this sequence.

To identify this soft direction, let n_A^{soft} be its normalized covector and let n_A^{ILove} be the I-Love plane normal embedded in the full observable space with zero compactness component. We define

$$\mathcal{A}_{\text{ILove}} = |n_A^{\text{soft}} g^{AB} n_B^{\text{ILove}}| \quad (55)$$

for unit covectors in the baseline observable metric. Along the reference sequence,

$$0.9924 \leq \mathcal{A}_{\text{ILove}} \leq 0.9958.$$

The full soft mode is therefore not an unrelated three-observable combination. It is nearly the I-Love normal, with a small compactness component. At $C = 0.115$, for example,

$$n^{\text{soft}} \simeq (0.099, -0.947, 0.304), \quad (56)$$

in the coordinate ordering $(\ln C, \ln \bar{I}, \ln \Lambda_2)$, while the embedded I-Love normal is approximately

$$(0, -0.957, 0.291).$$

The same structure persists when the expansion point is moved through the broad latent-field ensemble described below. Across the 44 stable backgrounds,

$$\text{median}(\mathcal{R}_N) = 0.0204, \quad q_{0.90} = 0.0469, \quad \max \mathcal{R}_N = 0.0883, \quad (57)$$

while the median I-Love alignment is 0.9977. Restricting to $C \geq 0.05$, the largest normal-mode ratio is 0.0797, the minimum alignment is 0.9756, and the median alignment is 0.9979. The principal geometric result is therefore that, after removing motion along the stellar sequence, the relativistic EOS response contains a distinguished soft normal mode that is robustly I-Love-like over large functional EOS deformations.

D. Broad nonparametric background ensemble

We next move the expansion point itself through EOS function space. Starting from the same low-density anchor, we draw 16 finite latent sound-speed backgrounds with unit RMS-scale perturbations: eight from a smooth squared-exponential process with $\ell_h = 0.08$ and eight from a rougher Matérn-3/2 process with $\ell_h = 0.04$. Each background remains causal by construction. The local response Jacobian is recomputed about the nonzero field at $h_c = 0.12, 0.20, 0.28$. Four of the 48 configurations lie on a negative- dM/dh_c branch and are excluded from the stable-star summary.

At the fine 2048-step stellar and 1024-point thermodynamic resolution, the 44 stable configurations have

$$\text{median} \left(\frac{\sigma_{\perp}^{\text{I Love}}}{\sigma_{\perp}^{\text{C Love}}} \right) = 0.0697, \quad q_{0.90} = 0.205. \quad (58)$$

All stable configurations retain an I-Love response smaller than C-Love; the largest ratio is 0.486. The two weakest cases occur at very low compactness, $C = 0.030$ and 0.045 . Once configurations with $C < 0.05$ are removed, all 41 remaining stable points satisfy

$$\frac{\sigma_{\perp}^{\text{I Love}}}{\sigma_{\perp}^{\text{C Love}}} < 0.231, \quad (59)$$

and for $C \geq 0.08$ the maximum is 0.218. The smooth and rough background families give similar median ratios, 0.064 and 0.072, respectively.

This result is deliberately broader than the small-polytrope study but should not be confused with a nuclear-physics posterior over realistic EOSs. It tests the functional geometry under large causal deformations of the sound-speed profile. Within that domain, the suppression is generic through the neutron-star compactness range and weakens most clearly toward the low-compactness limit.

A coarse 512/256 screening calculation initially assigned one rough background a response ratio 0.80. Recomputing the same configuration at the production resolution gives 0.218. This case demonstrates that response ratios can be substantially more resolution-sensitive than the baseline stellar observables for unusual EOS profiles. All ensemble statistics quoted above therefore use the fine calculation.

VI. INTEGRABILITY AND GLOBAL RECONSTRUCTION

A soft normal direction is a local statement. A common quasi-universal surface additionally requires the corresponding one-form to be approximately integrable. This is nontrivial only in the full three-dimensional observable space: a line field in the two-dimensional I-Love plane is locally integrable automatically.

A. Frobenius obstruction

At each stellar configuration we construct a three-dimensional domain patch

$$x^i = (h_c, \alpha_{\text{hard}}, \alpha_{\text{soft}}),$$

where the EOS directions are the covariance-normalized right singular vectors of the projected EOS response. Centered derivatives of both $y^A = (\ln C, \ln \bar{I}, \ln \Lambda_2)$ and the unit soft covector n_A give

$$\frac{\partial n_A}{\partial y^B} = \frac{\partial n_A}{\partial x^i} \left(\frac{\partial y}{\partial x} \right)^{-1 i}{}_B. \quad (60)$$

In the baseline Euclidean metric on logarithmic observables, the codimension-one Frobenius condition becomes

$$n \cdot (\nabla_y \times n) = 0. \quad (61)$$

We quote

$$\eta_F = \frac{|n \cdot (\nabla_y \times n)|}{|\nabla_y \times n|}. \quad (62)$$

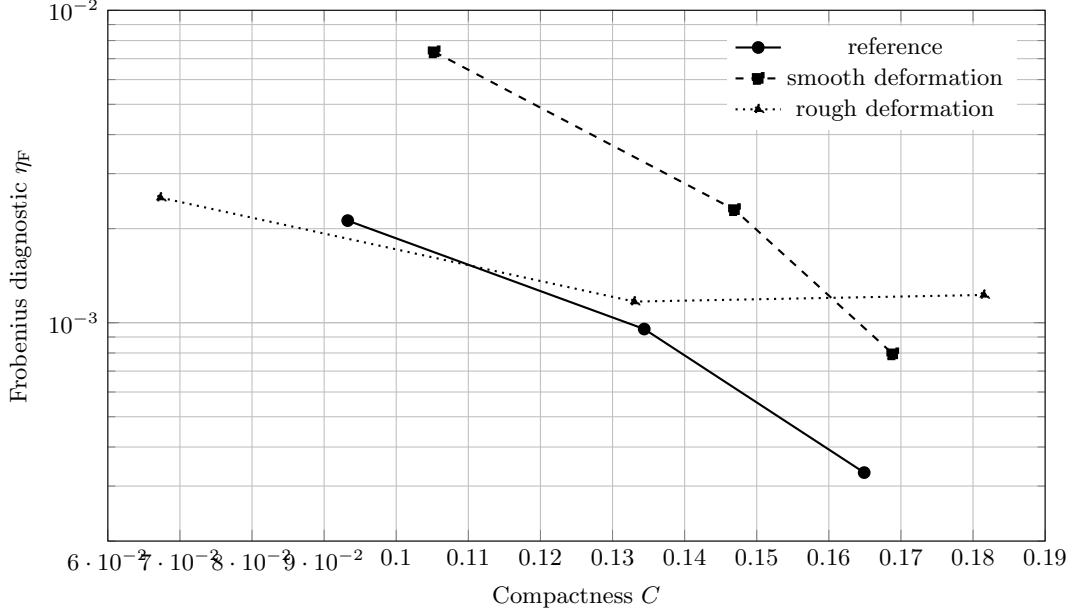


FIG. 2. High-resolution Frobenius obstruction of the soft normal field. The normalized obstruction remains at the 10^{-3} level across undeformed, smooth, and rough EOS backgrounds.

Because derivatives of the response eigenvector are more resolution-sensitive than the first-order observables, this calculation is performed with 4096 stellar steps and 2048 EOS reconstruction points. Across nine patches—the undeformed reference, one smooth finite EOS deformation, and one rough finite deformation, each at $h_c = 0.12, 0.20, 0.28$ —we find

$$\text{median}(\eta_F) = 1.23 \times 10^{-3}, \quad q_{0.90} = 3.48 \times 10^{-3}, \quad \max \eta_F = 7.36 \times 10^{-3}. \quad (63)$$

A candidate 3×10^{-2} obstruction seen at half this resolution collapses to 1.23×10^{-3} at the production second-derivative resolution, so we do not interpret the coarse value as physical non-integrability.

B. Cross-validated scalar potential

The local Frobenius result can be tested globally without fitting the stellar observables themselves. Since the $\ln \bar{I}$ component of the soft covector is nonzero throughout the sampled domain, normalize the one-form as

$$d \ln \bar{I} + A(\mathbf{y}) d \ln C + B(\mathbf{y}) d \ln \Lambda_2 \simeq 0. \quad (64)$$

If the field is integrable there exists a scalar

$$F(\mathbf{y}) = \ln \bar{I} + g(\ln C, \ln \Lambda_2) \quad (65)$$

with

$$\partial_{\ln C} g \simeq A, \quad \partial_{\ln \Lambda_2} g \simeq B.$$

We reconstruct g using only these gradient constraints; the observable values themselves are never regression targets.

The dataset contains 79 stable configurations with $C \geq 0.05$, drawn from 16 smooth and rough latent-field backgrounds at six central enthalpies. Polynomial potentials of degree two through four are tested. A cubic is retained as the reconstruction diagnostic because it improves over the quadratic in both background cross-validation and EOS-family transfer, whereas the quartic begins to overfit one family direction.

In four-fold cross-validation with complete EOS backgrounds held out, the cubic reconstruction gives

$$\text{median } \mathcal{A}(\nabla F, n_{\text{soft}}) = 0.9989 \quad (66)$$

TABLE IV. Median relative RMSE across nine background configurations for finite EOS displacement along random GP directions.

Amplitude s	Linear	Linear + quadratic
0.05	0.142	0.0062
0.10	0.285	0.024
0.25	0.650	0.156
0.50	1.028	0.553

and median held-out scatter

$$\sigma(F) = 8.48 \times 10^{-3}. \quad (67)$$

For a one-dimensional reconstruction constrained to depend only on $\ln \Lambda_2$, using the same gradient data,

$$\sigma(F_{I\text{Love}}) = 1.98 \times 10^{-2}.$$

The compactness-dependent soft surface therefore reduces the held-out scalar scatter by 57%.

The result transfers between qualitatively different EOS-background families. Training only on smooth backgrounds and testing only on rough ones gives

$$\sigma(F) = 8.89 \times 10^{-3}, \quad \sigma(F_{I\text{Love}}) = 2.60 \times 10^{-2},$$

while reversing the train/test families gives

$$\sigma(F) = 9.27 \times 10^{-3}, \quad \sigma(F_{I\text{Love}}) = 1.88 \times 10^{-2}.$$

These are reductions of 66% and 51%, respectively. Since the reconstruction is trained on the covector field rather than on the surface values, this probes the common integrability of the local soft directions rather than the flexibility of a direct fit through the stellar points.

Equations (63)–(67) provide complementary local and global tests of H3. We use the polynomial surface only as an integrability diagnostic here and do not interpret its particular functional form as a new fundamental universal relation.

VII. CURVATURE AND FINITE EOS SCATTER

We next test whether the leading finite departure from the *local soft tangent plane* is controlled by the second derivative of the EOS-to-observable map. At a fixed background configuration, let n_0 be the frozen soft covector and let d be a random unit-scale EOS perturbation drawn from the declared GP metric. Define the normal displacement

$$R(s) = n_0 \cdot [\mathbf{y}(a_0 + sd) - \mathbf{y}(a_0)]. \quad (68)$$

The first- and second-order predictions are

$$R_1(s) = s n_0 J d, \quad (69)$$

$$R_2(s) = s n_0 J d + \frac{s^2}{2} n_0 H[d, d]. \quad (70)$$

The directional curvature $n_0 H[d, d]$ is evaluated with a centered second difference at latent amplitude 0.03.

We use 48 random EOS directions at each of nine zero, smooth, and rough background configurations. Table IV summarizes the relative RMSE of the two approximations against the exact finite displacement.

At $s = 0.05$ and 0.10 , the second-order prediction is essentially quantitative, with 90th-percentile relative RMSE 0.016 and 0.058, respectively. At $s = 0.25$, the median error remains 0.156, while the curvature contribution itself carries a median 63% of the exact normal RMS. Thus the finite departure is already curvature dominated in a regime where the quadratic expansion remains predictive. By $s = 0.5$, the second-order approximation is no longer quantitatively reliable.

The curvature coefficient is stable to its finite-difference radius: changing the curvature step from 0.015 to 0.06 changes the directional-curvature RMS by 2.5×10^{-3} fractionally in the reference convergence test, with correlation > 0.99999 . These results support the local form of H4 over a finite but explicitly limited neighborhood of EOS function space: after the first-order soft response is suppressed, the extrinsic curvature of the EOS-to-observable image controls displacement away from the local soft tangent plane until higher orders become important. This test does not by itself evaluate the additional Hessian term associated with the curvature of the globally reconstructed scalar F .

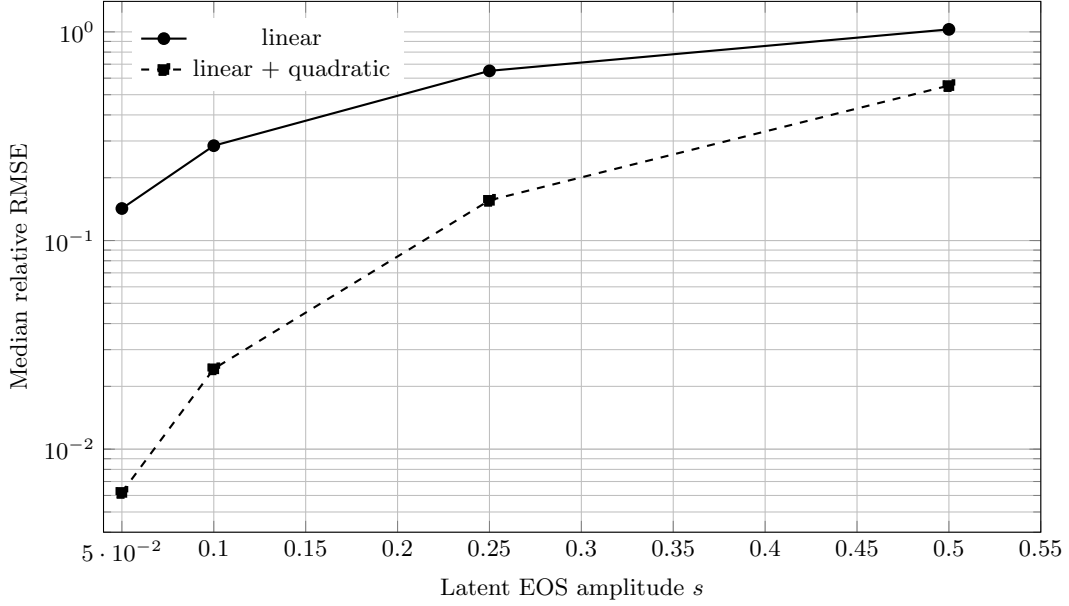


FIG. 3. Finite-radius accuracy of the local response expansion. Including the directional curvature reduces the prediction error by more than an order of magnitude for small EOS displacements and remains useful through $s \simeq 0.25$ before higher orders become important.

VIII. CONTROLLED BREAKDOWN

The broad latent-field ensemble above contains rough EOSs but does not isolate a localized structural feature. To test a controlled breakdown mechanism we therefore impose a smooth, phase-transition-like softening window directly in the latent sound-speed field. The sound speed remains causal and the thermodynamics are reconstructed exactly as in the baseline calculation. The profile is

$$\delta u(h) = -\frac{D}{2} \left[\tanh \frac{h - h_-}{\Delta h_{\text{edge}}} - \tanh \frac{h - h_+}{\Delta h_{\text{edge}}} \right], \quad (71)$$

where $h_{\pm}$ define the softened interval. We use 65 EOS nodes and 2048-point stellar and thermodynamic integrations for the refined cases.

For a window centered at $h = 0.16$, width 0.04, depth $D = 2$, and edge scale 0.008, the minimum sound speed squared is $c_s^2 \simeq 0.0198$. At $h_c = 0.30$ the compactness changes from $C = 0.1713$ in the undeformed star to $C = 0.1065$. More importantly, the response geometry changes from

$$\mathcal{A}_{I\text{Love}} : 0.9924 \rightarrow 0.9024, \quad (72)$$

$$\frac{\sigma_{\perp}^{I\text{Love}}}{\sigma_{\perp}^{C\text{Love}}} : 0.1348 \rightarrow 0.4956, \quad (73)$$

$$\mathcal{R}_N : 0.0517 \rightarrow 0.0962. \quad (74)$$

Thus the ordinary I-Love projection degrades by almost a factor of four in relative RMS response, while the full observable-space response still contains a pronounced soft normal direction. The effect is therefore better described as a rotation of the low-response manifold than as a disappearance of low-dimensional structure.

A broader window of width 0.08, evaluated at $h_c = 0.36$, gives the same qualitative result:

$$\mathcal{A}_{I\text{Love}} = 0.9260, \quad \frac{\sigma_{\perp}^{I\text{Love}}}{\sigma_{\perp}^{C\text{Love}}} = 0.4306, \quad \mathcal{R}_N = 0.1203.$$

The persistence of $\mathcal{R}_N \ll 1$ is important: a localized EOS feature can spoil the standard two-observable relation without destroying the hierarchy of the full EOS response.

The origin of the rotation is visible directly in the EOS sensitivity kernel. For the first transition case, the covariance-weighted I-Love normal variance increases from

$$5.99 \times 10^{-6} \quad \text{to} \quad 5.10 \times 10^{-4},$$

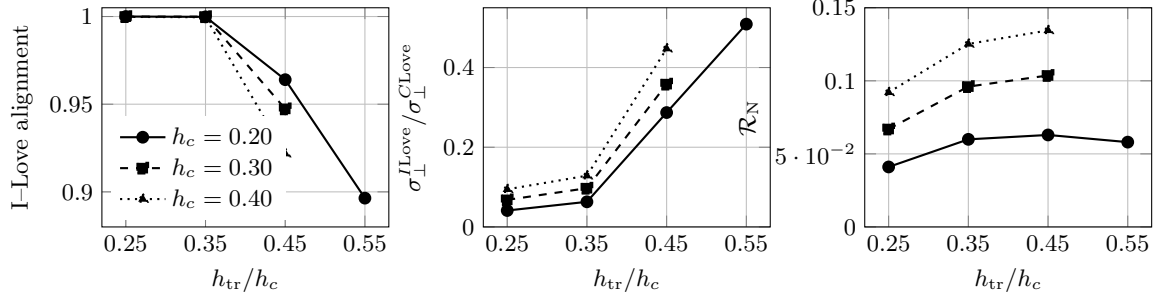


FIG. 4. High-resolution transition-location scan at fixed depth $D = 1.5$ and width $0.25h_c$. Only stable configurations passing the physical screen are shown. The I-Love projection can rotate strongly near $h_{tr}/h_c \simeq 0.45$ – 0.55 (left and center), while the full normal hierarchy remains separated (right).

a factor of 85.1. Of the positive excess in the direct nodal I-Love sensitivity, 90.2% lies inside the imposed interval $h = 0.14$ – 0.18 . The excess peaks at $h = 0.155$ and has centroid $h = 0.156$. The breakdown of the I-Love projection is therefore localized on the softened layer rather than being a global numerical displacement of the star.

This controlled example supports H5 in a more specific form than originally posed. Sharp EOS structure need not eliminate the low-response manifold. Instead, it can inject a localized EOS-sensitive direction into a particular projection, rotating the globally soft covector away from the ordinary I-Love normal. The systematic scan below tests how this rotation depends on the location, depth, and width of the softened layer.

A. Systematic transition-location scan

We scanned transition centers $h_{tr} = 0.10, 0.14, 0.18, 0.22$, widths 0.04 and 0.08, depths $D = 0.5, 1, 1.5, 2$, and three stellar configurations at $h_c = 0.20, 0.30, 0.40$. Throughout these scans, “stable” denotes the positive-mass-slope screen $dM/dh_c > 0$; we do not solve a separate radial-mode eigenvalue problem. Of the 96 screening configurations, 69 pass this screen and satisfy $0 < C < 0.35$. The strongest stable I-Love degradation moves with the stellar sequence rather than remaining at one absolute enthalpy: for fixed width 0.08 and depth 1.5, the extrema occur near

$$(h_c, h_{tr}) = (0.20, 0.10), (0.30, 0.14), (0.40, 0.18), \quad (75)$$

corresponding to $h_{tr}/h_c \simeq 0.45$ – 0.50 .

We refined these cases with 65 EOS nodes and 2048-point stellar and thermodynamic integrations. The response effect survives. The three refined I-Love/C-Love RMS ratios are 0.756, 0.439, and 0.388, while the corresponding full soft-to-hard RMS ratios remain 0.063, 0.106, and 0.126. More than 90% of the positive excess I-Love sensitivity remains localized on the softened interval in all three cases.

To separate the diffuse low-compactness endpoint from the more compact stars, we additionally scanned transition location in units of the central enthalpy, with fixed h -width $0.25h_c$ and depth $D = 1.5$. For configurations with $C \geq 0.1$, the strongest stable rotation occurs at $h_{tr}/h_c = 0.45$. At $h_c = 0.40$, this configuration has $C = 0.113$,

$$\mathcal{A}_{I\text{Love}} = 0.922, \quad \frac{\sigma_{\perp}^{I\text{Love}}}{\sigma_{\perp}^{C\text{Love}}} = 0.447, \quad \mathcal{R}_N = 0.135. \quad (76)$$

The I-Love variance is enhanced by a factor 56.7, and 93.3% of the positive excess direct sensitivity lies within the imposed layer.

The enthalpy-coordinate background profile provides a physical location for this vulnerability. In the deformed star the softened layer is centered at

$$r/R = 0.453, \quad m/M = 0.759, \quad (77)$$

and spans approximately $0.386 < r/R < 0.535$. The same enthalpy interval in the undeformed reference configuration is centered at $r/R = 0.618$. These numbers are properties of the controlled $\Gamma = 2$ experiment, not a universal radius.

A preliminary cross-EOS control was subsequently rejected by an explicit branch audit: one matched reference lay on a negative- dM/dh_c branch, and the first fixed-final-compactness depth solver assumed a globally monotonic compactness response that is not satisfied at all transition locations. Those intermediate values are retained in the project history but are not used for scientific claims here.

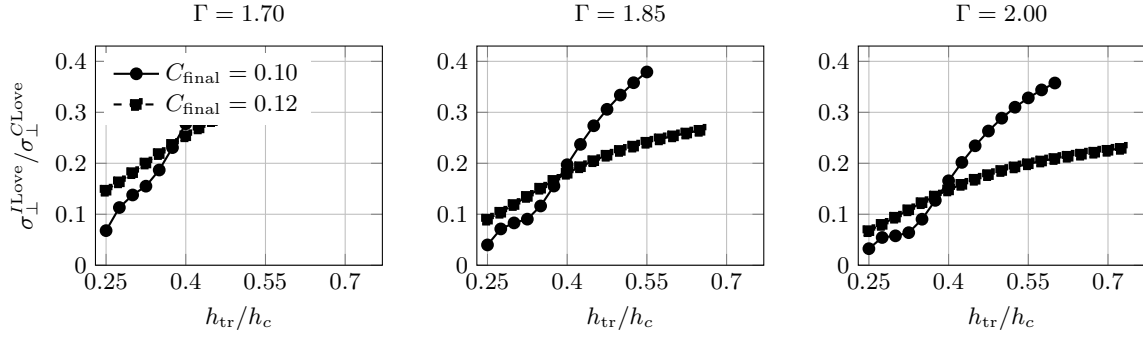


FIG. 5. Branch-safe fixed-final-compactness transition scan. The three reference polytropes start at the common baseline compactness $C = 0.14$ with positive dM/dh_c . At each transition location, the smallest-depth root connected continuously to the undeformed configuration is used to reach the specified final compactness; points are shown only while the positive-mass-slope screen remains satisfied along the deformation path. In every panel and for both target compactnesses, the I-Love/C-Love response grows toward the end of the retained branch. The largest retained response therefore tracks the branch boundary rather than selecting an interior, reference-EOS-independent transition shell.

B. Branch-safe cross-EOS control

We therefore repeated the comparison with a branch-safe construction. The three reference polytropes $\Gamma = 1.70, 1.85, 2.00$ are matched at the lower common baseline compactness $C = 0.14$, where their mass slopes are positive:

$$\frac{dM}{dh_c} = 0.707, 2.902, 2.685, \quad (78)$$

respectively. For each transition center $x \equiv h_{\text{tr}}/h_c$, with width $0.25h_c$, we sample the compactness as a function of softening depth over $0 \leq D \leq 3$, bracket all roots of $C(D) - C_{\text{final}}$, and select the smallest-depth root connected continuously to $D = 0$. We retain a point only if $dM/dh_c > 0$ at the target configuration and at 11 intermediate depths along the deformation path. The location grid covers $0.25 \leq x \leq 0.875$, the full range for which the nominal $0.25h_c$ -wide layer remains inside the star, and we test $C_{\text{final}} = 0.10$ and 0.12 .

The depth-root construction is numerically well behaved. The maximum compactness residual at a refined root is 1.1×10^{-15} . Four high- x cases possess two depth roots, demonstrating why a global monotonicity assumption is unsafe; all four fail the positive-mass-slope path screen and therefore do not enter the retained comparison.

The branch-safe result does not select an interior universal shell. For $C_{\text{final}} = 0.10$, the largest retained I-Love/C-Love response occurs at

$$x = 0.475, 0.550, 0.600$$

for $\Gamma = 1.70, 1.85, 2.00$, respectively. For $C_{\text{final}} = 0.12$, the corresponding locations are

$$x = 0.500, 0.650, 0.725.$$

In every one of these six cases, this is also the largest x that survives the positive-mass-slope path screen; the next grid point fails it. The response is still increasing over the final retained points. The apparent maximum therefore tracks the screened branch boundary rather than a preferred transition location.

Across the six boundary-adjacent retained points, the I-Love/C-Love RMS ratio spans 0.229–0.398, the I-Love alignment spans 0.939–0.979, and the full soft-to-hard RMS ratio remains 0.046–0.139. Mapping their layer centers into stellar coordinates gives

$$0.368 \leq r_{\text{tr}}/R \leq 0.476, \quad 0.235 \leq m_{\text{tr}}/M \leq 0.675.$$

Neither enthalpy coordinate nor physical shell coordinate collapses to a reference-EOS-independent value. The robust conclusion is therefore the localization capability demonstrated by the fixed-depth experiment, not a universal vulnerable shell.

IX. DISCUSSION

The response geometry separates three statements that are combined in an empirical universal relation. First, EOS perturbations may excite one normal observable-space direction much less than another. Second, the local soft

covectors must be approximately integrable to define a common global surface. Third, finite departures from the local soft tangent plane acquire a controlled nonlinear contribution once the first-order term is suppressed. The calculations above support these statements in those specific senses for $(C, \bar{I}, \Lambda_2)$ over the tested cold, causal EOS domain. The first statement is closely related to recent linear-response analyses of fixed universal relations [6] and to asymptotic reductions of the stellar equations [7]; the distinction here is that the weakly excited normal direction is inferred from the full metric-weighted response operator before a relation is chosen, and is then tested for integrability and nonlinear curvature.

The central result is stronger than the observation that the I–Love band is thin. After displacement along a stellar sequence is removed, the full three-observable response possesses a separated normal spectrum. Across the broad finite-background ensemble, the median soft-to-hard normal RMS ratio is 0.0204, with 90th percentile 0.0469. The corresponding covector is nearly the I–Love normal, with median alignment 0.9977, but it carries a small compactness component. This compactness dependence should not be confused with discovery of a new I–Love–C relation: analytic and numerical I–Love–C relations are already established [16]. Here compactness enters because the local low-response direction is inferred from the full EOS-response operator rather than chosen in advance.

The explicit EOS covariance makes one source of apparent universality transparent. Scatter measured over a finite EOS catalog implicitly assigns a distribution to EOS variations. In the present construction that weighting is exposed as C_{EOS} . The I–Love-like hierarchy survives changes in the covariance kernel, correlation length, nodal resolution, reference EOS, and observable normalization metric considered here. It is therefore not tied to one discrete set of named EOSs, although the numerical response scales remain metric dependent by construction.

The integrability result connects local stationarity to a global relation. The Frobenius obstruction remains at the 10^{-3} level over the high-resolution test patches, while a scalar potential inferred only from local gradient information generalizes to unseen EOS backgrounds. The potential is useful here as a diagnostic of the soft distribution, not as a preferred new fitting formula. In particular, the improvement over a one-dimensional I–Love reconstruction demonstrates that the compactness component of the local soft mode is coherent across EOSs rather than random noise.

The curvature calculation gives a finite radius to this geometric picture. For sufficiently small EOS displacements the quadratic response predicts the normal departure at percent-level accuracy, and at moderate amplitude the curvature term becomes the dominant contribution to the residual. At larger deformation the second-order expansion fails, as expected. This distinction is important: first-order stationarity does not imply negligible scatter for arbitrarily broad EOS priors.

The controlled softening experiment also changes the interpretation of “breakdown.” Sharp phase transitions are already known to degrade the accuracy of tidal universal relations [17], while useful universal behavior may persist in models with exotic or hybrid cores [18]. Our response calculation resolves these statements: a localized soft layer can rotate the low-response covector away from the ordinary I–Love projection without eliminating the separation of the full normal spectrum. In the refined example, the covariance-weighted I–Love variance increases by a factor 85, yet the soft-to-hard normal RMS ratio remains 0.096. More than 90% of the added direct I–Love sensitivity is localized on the imposed layer. The geometry therefore distinguishes failure of a chosen two-observable projection from failure of low-dimensional EOS response itself.

The transition-location experiment should not be turned into a universal-shell claim. The fixed-depth $\Gamma = 2$ scan shows that the response formalism can localize the EOS perturbation responsible for rotating the soft mode. A branch-safe cross-EOS control, constructed only from positive-mass-slope baselines and first depth roots connected to the undeformed configuration, does not reveal a common interior maximum. Instead, for every tested reference EOS and both final compactnesses, the I–Love/C–Love response continues to rise until the positive-mass-slope path screen truncates the sequence. The location of that boundary shifts from $h_{\text{tr}}/h_c = 0.475$ to 0.725, with corresponding layer centers spanning $r/R = 0.368$ –0.476 and $m/M = 0.235$ –0.675. Thus the robust result is the ability to localize a response vulnerability for a specified stellar/EOS configuration, not the existence of a reference-EOS-independent vulnerable shell.

This density-resolved behavior connects naturally with earlier physical accounts of I–Love–Q universality based on self-similarity, stiffness, and the dominant contribution of the stellar outer layers [2, 3]. The present framework does not replace those explanations. It supplies a differential diagnostic that can locate which EOS perturbations excite the otherwise soft observable direction and can test whether the resulting rotated distribution remains integrable.

Several limitations are deliberate. We consider cold barotropic stars in general relativity, static tides, and first-order slow rotation. The broad latent-field ensemble is a function-space stress test rather than a nuclear-physics posterior. The transition profiles are controlled softening experiments rather than full microphysical first-order phase-transition constructions, and the positive- dM/dh_c screen used in the transition scans is not a direct radial-mode stability calculation. Finally, observable-space norms require a declared metric; only statements invariant under coordinate changes with the metric transformed consistently are geometric.

X. CONCLUSIONS

We have recast neutron-star quasi-universal relations as low-response structure of the relativistic EOS-to-observable map. With an explicit metric on EOS function space, removal of the stellar-sequence tangent exposes a strongly separated soft normal mode in $(\ln C, \ln \tilde{I}, \ln \Lambda_2)$. The mode is robustly I-Love-like but not exactly confined to the I-Love plane. Its covector field is approximately integrable across large causal EOS deformations, and the corresponding scalar potential generalizes across held-out EOS backgrounds when reconstructed from gradients alone. Second-order curvature quantitatively predicts the initial finite departure from the local soft tangent plane.

Localized EOS softening provides the complementary breakdown test. It can rotate the soft covector enough to substantially degrade I-Love while the full normal hierarchy survives, and the excess sensitivity is localized on the responsible stellar layer. A branch-safe cross-EOS stress test shows that the strongest retained degradation follows the screened stability boundary rather than a universal stellar shell. Quasi-universality is therefore usefully described not only by the scatter of a fitted relation, but by the rank, orientation, integrability, and curvature of the EOS response itself.

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